Double Integrals in polar Coordinates

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The Gaussian Integral // Solved Using Polar Coordinates

Use the following online solver:

Polar/Rectangular Coordinates Calculator - eMathHelp

Polar/Rectangular Equation Calculator - eMathHelp

Polar equation plotter – GeoGebra

PRACTISE: HOWARD ANTON: 14.3: Double Integrals in polar Coordinates: 1-12, 23-34

1–6 Evaluate the iterated integral. ■

1. $\displaystyle\int_0^{\pi/2}\int_0^{\sin\theta} r\cos\theta\, dr\, d\theta$

2. $\displaystyle\int_0^{\pi}\int_0^{1+\cos\theta} r\, dr\, d\theta$

3. $\displaystyle\int_0^{\pi/2}\int_0^{a\sin\theta} r^2\, dr\, d\theta$

4. $\displaystyle\int_0^{\pi/6}\int_0^{\cos 3\theta} r\, dr\, d\theta$

5. $\displaystyle\int_0^{\pi}\int_0^{1-\sin\theta} r^2\cos\theta\, dr\, d\theta$

6. $\displaystyle\int_0^{\pi/2}\int_0^{\cos\theta} r^3\, dr\, d\theta$

7–10 Use a double integral in polar coordinates to find the area of the region described. ■

7. The region enclosed by the cardioid $r = 1 - \cos\theta$.

8. The region enclosed by the rose $r = \sin 2\theta$.

9. The region in the first quadrant bounded by $r = 1$ and $r = \sin 2\theta$, with $\pi/4 \le \theta \le \pi/2$.

10. The region inside the circle $x^2 + y^2 = 4$ and to the right of the line $x = 1$.

FOCUS ON CONCEPTS

11–12 Let R be the region described. Sketch the region R and fill in the missing limits of integration.

$$\iint_R f(r,\theta)\,dA = \int_{\square}^{\square}\int_{\square}^{\square} f(r,\theta)r\,dr\,d\theta$$

11. The region inside the circle $r = 4\sin\theta$ and outside the circle $r = 2$.

12. The region inside the circle $r = 1$ and outside the cardioid $r = 1 + \cos\theta$.

23–26 Use polar coordinates to evaluate the double integral.

23. $\iint_R \sin(x^2 + y^2)\, dA$, where R is the region enclosed by the circle $x^2 + y^2 = 9$.

24. $\iint_R \sqrt{9 - x^2 - y^2}\, dA$, where R is the region in the first quadrant within the circle $x^2 + y^2 = 9$.

25. $\iint_R \frac{1}{1 + x^2 + y^2}\, dA$, where R is the sector in the first quadrant bounded by $y = 0$, $y = x$, and $x^2 + y^2 = 4$.

26. $\iint_R 2y\, dA$, where R is the region in the first quadrant bounded above by the circle $(x - 1)^2 + y^2 = 1$ and below by the line $y = x$.

27–34 Evaluate the iterated integral by converting to polar coordinates. ■

27. $\displaystyle\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2 + y^2)\, dy\, dx$

28. $\displaystyle\int_{-2}^{2} \int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}} e^{-(x^2+y^2)}\, dx\, dy$

29. $\displaystyle\int_0^2 \int_0^{\sqrt{2x-x^2}} \sqrt{x^2 + y^2}\, dy\, dx$

30. $\displaystyle\int_0^1 \int_0^{\sqrt{1-y^2}} \cos(x^2 + y^2)\, dx\, dy$

31. $\displaystyle\int_0^a \int_0^{\sqrt{a^2-x^2}} \frac{dy\, dx}{(1 + x^2 + y^2)^{3/2}} \quad (a > 0)$

32. $\displaystyle\int_0^1 \int_y^{\sqrt{y}} \sqrt{x^2 + y^2}\, dx\, dy$

33. $\displaystyle\int_0^{\sqrt{2}} \int_y^{\sqrt{4-y^2}} \frac{1}{\sqrt{1 + x^2 + y^2}}\, dx\, dy$

34. $\displaystyle\int_{-4}^{0} \int_{-\sqrt{16-x^2}}^{\sqrt{16-x^2}} 3x\, dy\, dx$